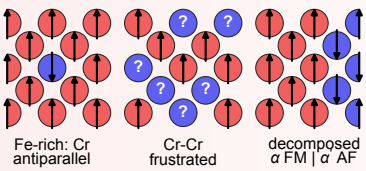
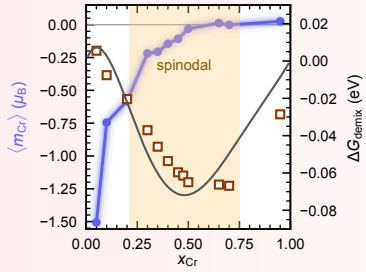
## FRUSTRATE

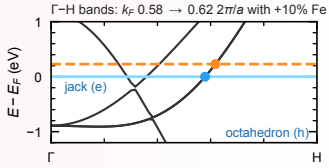
(a) atoms & moments

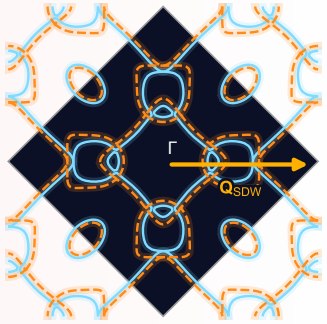



## NEST

(b) Cr Fermi surface vs Fe content

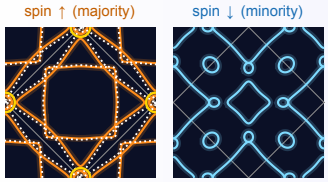


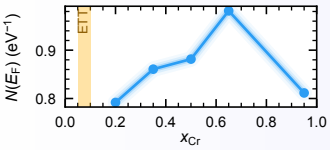

## TOPOLOGY

(c) Fe-rich ETT: spin ↑ vs ↓



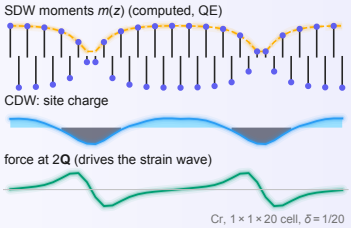




Cr brings the majority H level to $E_F$ ($0.10 \rightarrow 0.005$ eV at 5% Cr, exp. *a*) ⇒ ETT at $x_c \approx 0.08$, spin ↑ only (S6; Korzhavyi: 5–10%)



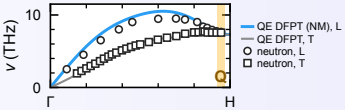

## MODULATE

(d) SDW couples to volume, not shear

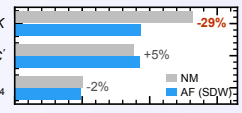



$P_{mag}$ vs NM: AF 5.8, SDW 4.2 GPa
SDW/AF 0.73 ≈ $\langle m^2 \rangle$ ratio 0.71 (Landau)
strain wave $u \approx 0.02$ Å, $\varepsilon \approx 0.5\%$, period 29 Å: averages out over a 6.4 μm SAW

Kohn dip near **Q**: zone-boundary L phonons only; $C_{ij}$ = slopes at Γ (NM too steep vs AF neutron data)



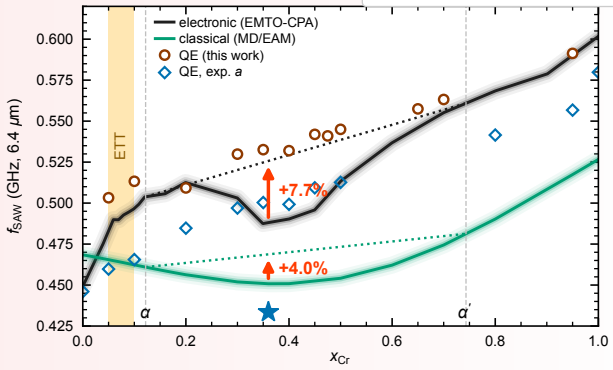

AF order: $K$ −29%, $f_{SAW}$ −1.0% (shear unchanged)




## STIFFEN

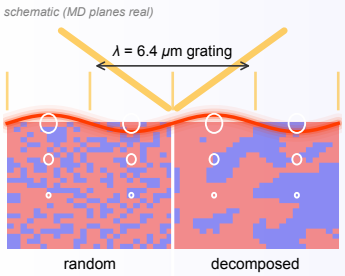
(e) electron → lattice: decomposition

$C = C_{ion} + C_{band}$, $C_{band} \propto -\sum_k (\partial\varepsilon_k/\partial\eta)^2 \delta(\varepsilon_k - E_F)$




## PROPAGATE

(f) Rayleigh wave in TGS




$C_{ij}\uparrow \Rightarrow G\uparrow \Rightarrow v_R = \eta(\nu)\sqrt{G/\rho}\uparrow \Rightarrow f = v_R/\lambda\uparrow$
Fe−36Cr, MD: 451 → 473 MHz (+5.0%, 5 alloys)
electronic lever rule (EMTO−CPA): +7.7%
QE lever rule: +0.2% (relaxed), −1.0% (exp. *a*): no gain

## DETECT

(g) (P)SAW vs pristine CASS

